M54564P/FP

8-UNIT 500mA SOURCE TYPE DARLINGTON TRANSISTOR ARRAY

DESCRIPTION

M54564P and M54564FP are eight-circuit output-sourcing Darlington transistor arrays. The circuits are made of PNP and NPN transistors. Both the semiconductor integrated circuits perform high-current driving with extremely low input-current supply.

FEATURES

- High breakdown voltage ($BV_{CEO} \geq 50V$)
- High-current driving (Io(max) = –500mA)
- With output pulldown resistance (Driving available with fluorescent display tube)
- Driving available with PMOS IC output or with TTL output
- Wide operating temperature range (Ta = –20 to +75°C)
- Output current-sourcing type

APPLICATION

Drives of relays, printers, LEDs, fluorescent display tubes and lamps, and interfaces between MOS-bipolar logic systems and relays, solenoids, or small motors

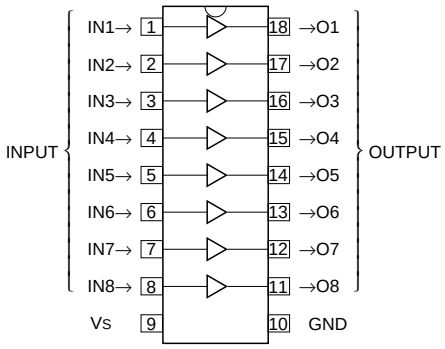
FUNCTION

The M54564P and M54564FP each have eight circuits, which are made of input inverters and current-sourcing outputs. The outputs are made of PNP transistors and NPN Darlington transistors. The PNP transistor base current is constant. Resistance of 50kΩ is connected between each output pin and GND, making these ICs suitable for fluorescent display tubes. Vs and GND are used commonly among the eight circuits.

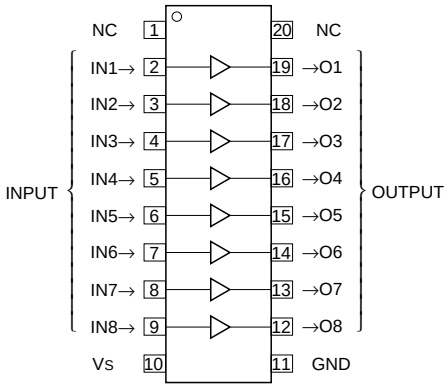
Output current is 500mA maximum. Supply voltage Vs is 50V maximum.

The M54564FP is enclosed in a molded small flat package, enabling space-saving design.

PIN CONFIGURATION



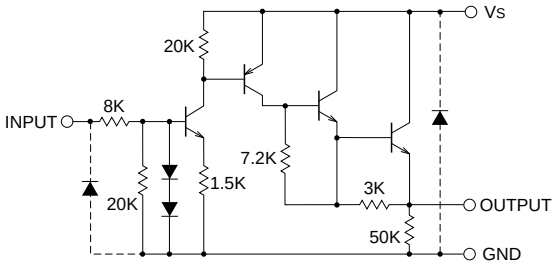
Package type 18P4G(P)



Package type 20P2N-A(FP)

NC : No connection

CIRCUIT DIAGRAM



The eight circuits share the Vs and GND.

The diode, indicated with the dotted line, is parasitic, and cannot be used.

Unit : Ω

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8-UNIT 500mA SOURCE TYPE DARLINGTON TRANSISTOR ARRAY

ABSOLUTE MAXIMUM RATINGS (Unless otherwise noted, Ta = –20 ~ +75°C)

Symbol	Parameter	Conditions	Ratings	Unit
V_{CEO}	Collector-emitter voltage	Output, L	–0.5 ~ +50	V
V_S	Supply voltage		50	V
V_I	Input voltage		–0.5 ~ +30	V
I_O	Output current	Current per circuit output, H	–500	mA
P_d	Power dissipation	Ta = 25°C, when mounted on board	1.79(P)/1.10(FP)	W
T_{opr}	Operating temperature		–20 ~ +75	°C
T_{stg}	Storage temperature		–55 ~ +125	°C

RECOMMENDED OPERATING CONDITIONS (Unless otherwise noted, Ta = –20 ~ +75°C)

Symbol	Parameter		Limits			Unit
			min	typ	max	
V_S	Supply voltage		0	—	50	V
I_O	Output current (Current per 1 circuit when 8 circuits are coming on simultaneously)	Duty Cycle P : no more than 8% FP : no more than 5%	0	—	–350	mA
		Duty Cycle P : no more than 55% FP : no more than 30%	0	—	–100	
V_{IH}	"H" input voltage		2.4	—	25	V
V_{IL}	"L" input voltage		0	—	0.2	V

ELECTRICAL CHARACTERISTICS (Unless otherwise noted, Ta = –20 ~ +75°C)

Symbol	Parameter	Test conditions	Limits			Unit
			min	typ*	max	
$I_{S\ (leak)}$	Supply leak current	V_S = 50V, V_I = 0.2V	—	—	100	μA
$V_{CE\ (sat)}$	Collector-emitter saturation voltage	V_S = 10V, V_I = 2.4V, I_O = –350mA	—	1.6	2.4	V
		V_S = 10V, V_I = 2.4V, I_O = –100mA	—	1.45	2.0	
I_I	Input current	V_I = 5V, V_S = 10V	—	0.4	0.7	mA
		V_I = 25V, V_S = 30V	—	2.9	6.5	
I_S	Supply current	V_S = 50V, V_I = 5V	—	—	5.0	mA

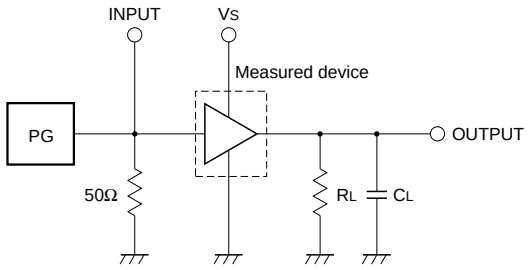
* : The typical values are those measured under ambient temperature (Ta) of 25°C. There is no guarantee that these values are obtained under any conditions.

SWITCHING CHARACTERISTICS (Unless otherwise noted, Ta = 25°C)

Symbol	Parameter	Test conditions	Limits			Unit
			min	typ	max	
t_{on}	Turn-on time	C_L = 15pF (note 1)	—	185	—	ns
t_{off}	Turn-off time		—	4300	—	ns

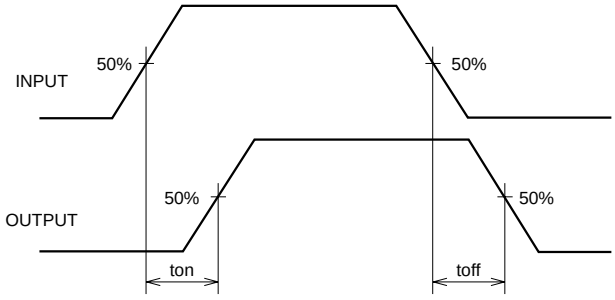
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NOTE 1 TEST CIRCUIT



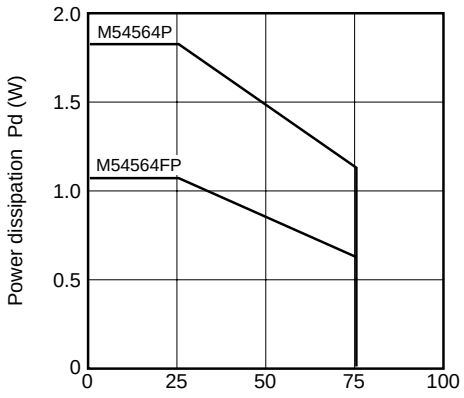
(1) Pulse generator (PG) characteristics : PRR = 1kHz,
tw = 10µs, tr = 6ns, tf = 6ns, ZO = 50Ω
VI = 0 to 2.4V
(2) Input-output conditions : RL = 30Ω, VS = 10V
(3) Electrostatic capacity CL includes floating capacitance at connections and input capacitance at probes

TIMING DIAGRAM

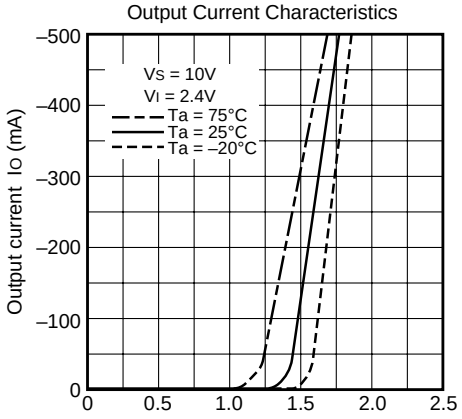


TYPICAL CHARACTERISTICS

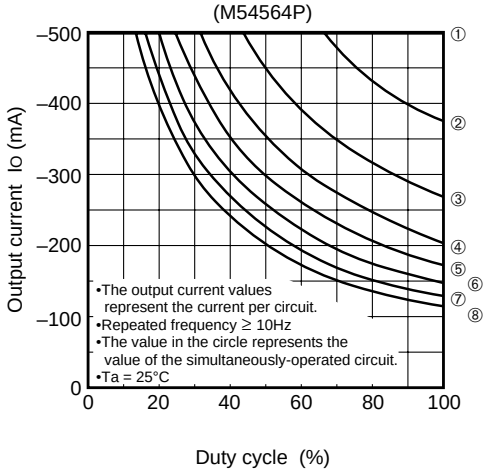
Thermal Derating Factor Characteristics



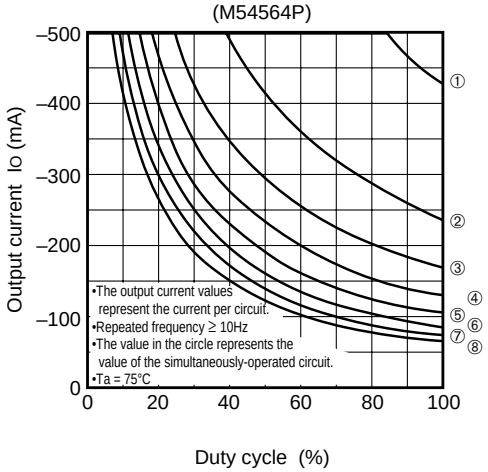
Output Saturation Voltage
Output Current Characteristics



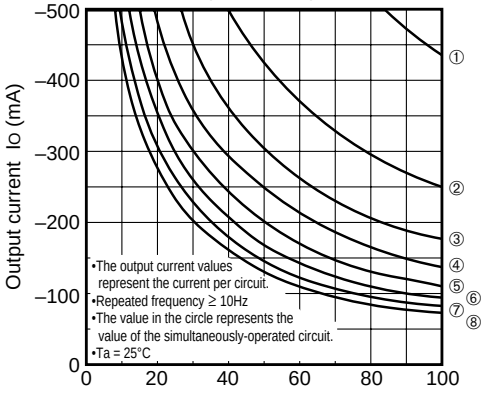
Duty-Cycle-Output Current Characteristics
(M54564P)



Duty-Cycle-Output Current Characteristics
(M54564P)

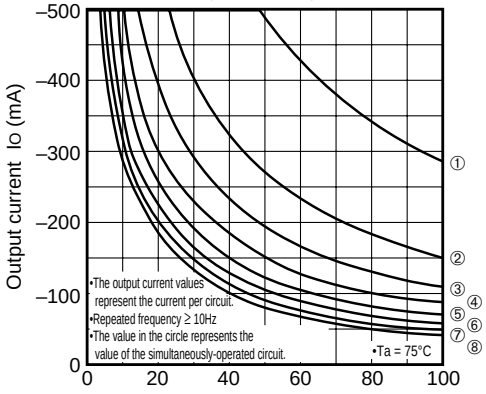


Duty-Cycle-Output Current Characteristics
(M54564FP)



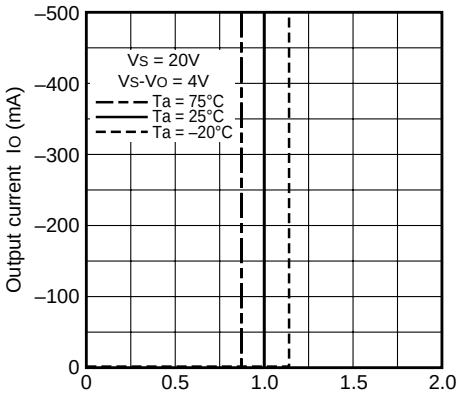
Duty cycle (%)

Duty-Cycle-Output Current Characteristics
(M54564FP)



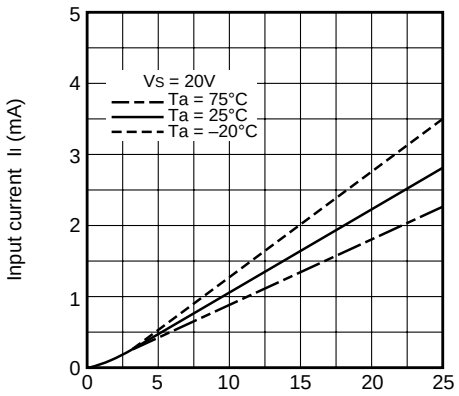
Duty cycle (%)

Grounded Emitter Transfer Characteristics



Input voltage VI (V)

Input Characteristics



Input voltage VI (V)